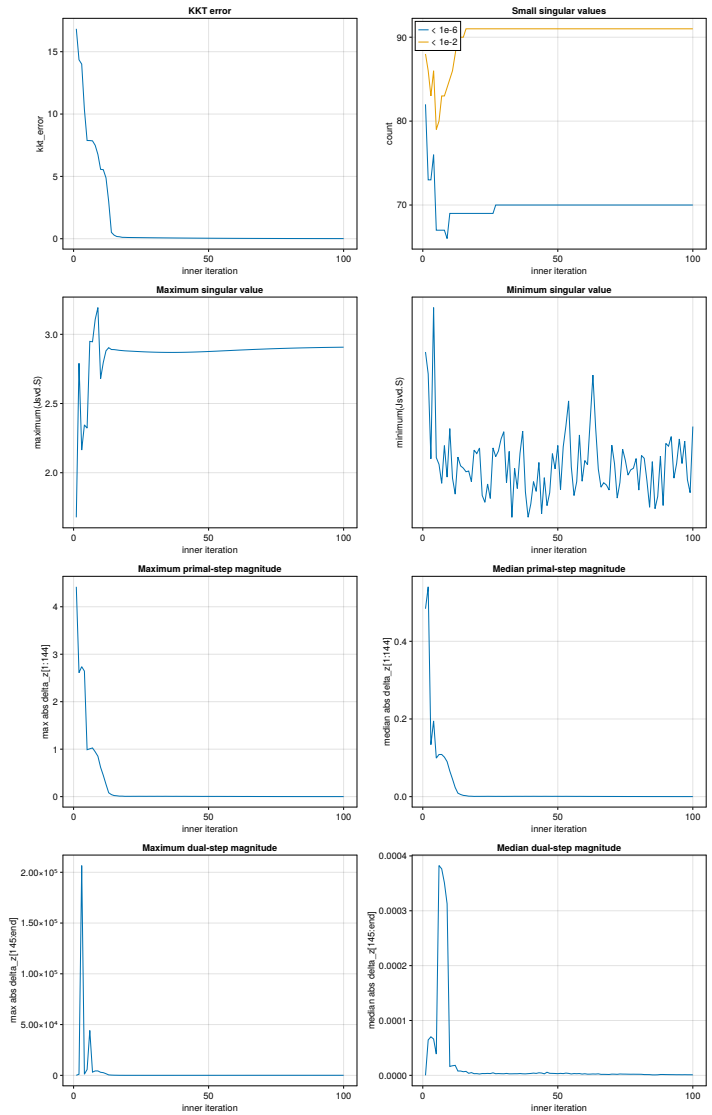
Solver Diagnostics

Recorded 100 inner iterations; reporting 100 rows (cap: 1000).

Summary statistics

Metric	Minimum	Maximum	Mean	Median
`kkt_error`	1.145437e-02	1.683624e+01	1.168830e+00	4.071875e-02
`count(Jsvd.S < 1e-6)`	6.600000e+01	8.200000e+01	6.991000e+01	7.000000e+01
`count(Jsvd.S < 1e-2)`	7.900000e+01	9.100000e+01	9.016000e+01	9.100000e+01
`maximum(Jsvd.S)`	1.677692e+00	3.193299e+00	2.858668e+00	2.884052e+00
`minimum(Jsvd.S)`	2.661519e-17	8.691072e-17	4.200173e-17	4.125054e-17
`maximum(abs.(delta_z[1:144]))`	9.224242e-04	4.420306e+00	1.906030e-01	4.789998e-03
`median(abs.(delta_z[1:144]))`	1.497935e-04	5.402795e-01	2.064208e-02	7.254584e-04
`maximum(abs.(delta_z[145:end]))`	1.220091e+01	2.066177e+05	2.817691e+03	4.204056e+01
`median(abs.(delta_z[145:end]))`	1.254924e-17	3.824504e-04	1.957083e-05	2.872102e-06

Metrics versus inner iteration



Per-iteration diagnostics

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max\lvert\delta z_{1:144}\rvert$	$\mathrm{med}\lvert\delta z_{1:144}\rvert$	$\max\lvert\delta z_{145:}\rvert$	$\mathrm{med}\lvert\delta z_{145:}\rvert$
1	1.683624e+01	82	88	1.677692e+00	7.413368e-17	4.420306e+00	4.838768e-01	2.421780e+01	1.254924e-17
2	1.435187e+01	73	86	2.790723e+00	6.781620e-17	2.611171e+00	5.402795e-01	1.114098e+03	6.444524e-05
3	1.400582e+01	73	83	2.163701e+00	4.343961e-17	2.737386e+00	1.337795e-01	2.066177e+05	7.026778e-05
4	1.035957e+01	76	86	2.345538e+00	8.691072e-17	2.644782e+00	1.950189e-01	1.133368e+03	6.625775e-05
5	7.877808e+00	67	79	2.322248e+00	4.377758e-17	9.884946e-01	9.916756e-02	5.757475e+03	3.873773e-05
6	7.873608e+00	67	80	2.948698e+00	4.185454e-17	1.007890e+00	1.084919e-01	4.422152e+04	3.824504e-04
7	7.850638e+00	67	83	2.946288e+00	3.639403e-17	1.025277e+00	1.085817e-01	3.197805e+03	3.764374e-04
8	7.490377e+00	67	83	3.107685e+00	4.723874e-17	9.439207e-01	1.017282e-01	4.350766e+03	3.510368e-04
9	6.760623e+00	66	84	3.193299e+00	3.817547e-17	8.508784e-01	9.016852e-02	4.332970e+03	3.127030e-04
10	5.565860e+00	69	85	2.679758e+00	5.212569e-17	6.171516e-01	6.592838e-02	3.008966e+03	1.625398e-05
11	5.538047e+00	69	86	2.797376e+00	3.787751e-17	4.480311e-01	4.528544e-02	2.650160e+03	1.747056e-05
12	4.879773e+00	69	88	2.880896e+00	3.328219e-17	2.593596e-01	2.344919e-02	1.609083e+03	1.818098e-05
13	2.973293e+00	69	90	2.903500e+00	4.393138e-17	8.161944e-02	9.037528e-03	4.296451e+02	7.840093e-06
14	5.089501e-01	69	90	2.890126e+00	4.141077e-17	4.253364e-02	5.552145e-03	2.467661e+02	7.879700e-06
15	2.880710e-01	69	90	2.889595e+00	4.075894e-17	2.283343e-02	3.237081e-03	1.359555e+02	6.603644e-06
16	1.777073e-01	69	91	2.886953e+00	3.972056e-17	1.764616e-02	2.434725e-03	1.006275e+02	7.172432e-06
17	1.581537e-01	69	91	2.884168e+00	3.992212e-17	1.026385e-02	1.208892e-03	4.421183e+01	3.705357e-06
18	1.126885e-01	69	91	2.882417e+00	3.686405e-17	1.041910e-02	1.045599e-03	4.099888e+01	4.967710e-06
19	1.088753e-01	69	91	2.880663e+00	4.593930e-17	6.244265e-03	6.325087e-04	2.103698e+01	3.030144e-06
20	1.007090e-01	69	91	2.879626e+00	4.494768e-17	6.713708e-03	7.349808e-04	2.140250e+01	3.129400e-06
21	9.808115e-02	69	91	2.878536e+00	4.653806e-17	6.372256e-03	7.135505e-04	1.873914e+01	2.404795e-06
22	9.518248e-02	69	91	2.877521e+00	3.292477e-17	6.972596e-03	7.645699e-04	1.905004e+01	3.283281e-06
23	9.267707e-02	69	91	2.876445e+00	3.093344e-17	6.681181e-03	7.701340e-04	1.665464e+01	3.217210e-06
24	8.980976e-02	69	91	2.875465e+00	3.619981e-17	7.349323e-03	8.755369e-04	1.691183e+01	3.491474e-06
25	8.742104e-02	69	91	2.874450e+00	3.204242e-17	7.049703e-03	8.183857e-04	1.475784e+01	3.211784e-06
26	8.457881e-02	69	91	2.873550e+00	4.657541e-17	6.762101e-03	7.534738e-04	1.357466e+01	4.552871e-06
27	8.199573e-02	70	91	2.872762e+00	4.406614e-17	6.487654e-03	6.998410e-04	1.263303e+01	2.932963e-06
28	7.963083e-02	70	91	2.872076e+00	4.566363e-17	7.132014e-03	7.631205e-04	1.358118e+01	3.233403e-06
29	7.759050e-02	70	91	2.871395e+00	4.916681e-17	6.849863e-03	7.045443e-04	1.258679e+01	2.997402e-06
30	7.520644e-02	70	91	2.870825e+00	5.122639e-17	6.587023e-03	6.431742e-04	1.220091e+01	2.869042e-06
31	7.301548e-02	70	91	2.870359e+00	3.654669e-17	7.264556e-03	7.221784e-04	1.401381e+01	3.476429e-06
32	7.113596e-02	70	91	2.869928e+00	4.557297e-17	6.940876e-03	7.571489e-04	1.399466e+01	2.768018e-06
33	6.891644e-02	70	91	2.869606e+00	2.661519e-17	6.620294e-03	7.649751e-04	1.392229e+01	2.793407e-06
34	6.687255e-02	70	91	2.869382e+00	4.069359e-17	7.240019e-03	8.645563e-04	1.590555e+01	2.932593e-06
35	6.513234e-02	70	91	2.869223e+00	3.489722e-17	6.863107e-03	8.335046e-04	1.581924e+01	2.916962e-06
36	6.305471e-02	70	91	2.869164e+00	4.505571e-17	6.498675e-03	8.358723e-04	1.569296e+01	3.276678e-06
37	6.113899e-02	70	91	2.869192e+00	5.140444e-17	6.155512e-03	7.933636e-04	1.556640e+01	2.835261e-06
38	5.936517e-02	70	91	2.869296e+00	3.394697e-17	6.683417e-03	8.849748e-04	1.776628e+01	2.685223e-06
39	5.782288e-02	70	91	2.869489e+00	2.664609e-17	6.292225e-03	8.390587e-04	1.766241e+01	3.092760e-06
40	5.600647e-02	70	91	2.869755e+00	3.047928e-17	5.920854e-03	8.262818e-04	1.753512e+01	3.436846e-06
41	5.432229e-02	70	91	2.870083e+00	3.688284e-17	6.383648e-03	9.087050e-04	2.005283e+01	4.062082e-06
42	5.288324e-02	70	91	2.870518e+00	3.407325e-17	5.970357e-03	8.576028e-04	1.998052e+01	3.514023e-06
43	5.115822e-02	70	91	2.871008e+00	4.238813e-17	5.582960e-03	8.176801e-04	1.988793e+01	4.633403e-06
44	4.956012e-02	70	91	2.871544e+00	2.763879e-17	5.226275e-03	7.914730e-04	1.981526e+01	4.201465e-06
45	4.807541e-02	70	91	2.872115e+00	3.801813e-17	5.604019e-03	8.724313e-04	2.276857e+01	2.766428e-06
46	4.679719e-02	70	91	2.872799e+00	2.997527e-17	5.214053e-03	8.383492e-04	2.278552e+01	5.613075e-06
47	4.527090e-02	70	91	2.873511e+00	3.371464e-17	4.852350e-03	8.031243e-04	2.277638e+01	3.670903e-06
48	4.385295e-02	70	91	2.874241e+00	4.493065e-17	4.673613e-03	7.692557e-04	2.278456e+01	3.546671e-06
49	4.253274e-02	70	91	2.874982e+00	4.053304e-17	5.173481e-03	8.511764e-04	2.629701e+01	3.364892e-06
50	4.139866e-02	70	91	2.875837e+00	4.727526e-17	4.946686e-03	8.116305e-04	2.641658e+01	3.069947e-06
51	4.003884e-02	70	91	2.876695e+00	3.454340e-17	4.727645e-03	7.754826e-04	2.649048e+01	3.573028e-06
52	3.877389e-02	70	91	2.877549e+00	4.686876e-17	4.516242e-03	7.417601e-04	2.657086e+01	3.138763e-06
53	3.759512e-02	70	91	2.878394e+00	5.287476e-17	4.954916e-03	8.172915e-04	3.074336e+01	4.067121e-06
54	3.659385e-02	70	91	2.879349e+00	6.007929e-17	4.698066e-03	7.753139e-04	3.094035e+01	3.536177e-06
55	3.537869e-02	70	91	2.880287e+00	4.109031e-17	4.455052e-03	7.362909e-04	3.106182e+01	2.542839e-06
56	3.424807e-02	70	91	2.881204e+00	3.287864e-17	4.224886e-03	6.998557e-04	3.117410e+01	3.234950e-06
57	3.319458e-02	70	91	2.882098e+00	3.686428e-17	4.007336e-03	6.658451e-04	3.127629e+01	2.973406e-06
58	3.220978e-02	70	91	2.882969e+00	5.029583e-17	4.364698e-03	7.287384e-04	3.617805e+01	3.238012e-06
59	3.137101e-02	70	91	2.883937e+00	3.699221e-17	4.110310e-03	6.801503e-04	3.637334e+01	2.312802e-06
60	3.035300e-02	70	91	2.884873e+00	4.294833e-17	3.873354e-03	6.364727e-04	3.645867e+01	2.875161e-06
61	2.940368e-02	70	91	2.885776e+00	4.173441e-17	3.652139e-03	5.987149e-04	3.651726e+01	2.139921e-06
62	2.851743e-02	70	91	2.886647e+00	5.451855e-17	3.445736e-03	5.682505e-04	3.655065e+01	2.204282e-06

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max\lvert\delta z_{1:144}\rvert$	$\mathrm{med}\lvert\delta z_{1:144}\rvert$	$\max\lvert\delta z_{145:}\rvert$	$\mathrm{med}\lvert\delta z_{145:}\rvert$
63	2.768775e-02	70	91	2.887487e+00	6.750521e-17	3.253204e-03	5.398656e-04	3.655953e+01	2.501632e-06
64	2.690908e-02	70	91	2.888295e+00	5.247920e-17	3.527232e-03	5.926818e-04	4.213834e+01	2.317306e-06
65	2.624030e-02	70	91	2.889186e+00	4.046849e-17	3.307567e-03	5.559303e-04	4.218632e+01	2.746701e-06
66	2.543219e-02	70	91	2.890038e+00	3.527109e-17	3.105044e-03	5.203186e-04	4.210110e+01	1.749573e-06
67	2.467617e-02	70	91	2.890854e+00	3.656155e-17	2.917753e-03	4.987187e-04	4.198002e+01	1.796713e-06
68	2.396828e-02	70	91	2.891635e+00	3.591090e-17	2.744471e-03	4.719740e-04	4.182717e+01	1.699464e-06
69	2.330390e-02	70	91	2.892382e+00	3.446340e-17	2.584053e-03	4.443058e-04	4.164542e+01	1.546368e-06
70	2.267902e-02	70	91	2.893097e+00	4.735670e-17	2.794530e-03	4.807458e-04	4.776231e+01	2.204819e-06
71	2.214653e-02	70	91	2.893881e+00	4.203788e-17	2.614032e-03	4.504246e-04	4.755681e+01	2.112254e-06
72	2.149608e-02	70	91	2.894626e+00	3.214867e-17	2.448617e-03	4.276212e-04	4.721388e+01	1.833753e-06
73	2.088646e-02	70	91	2.895336e+00	3.662407e-17	2.296474e-03	4.064392e-04	4.683876e+01	2.442949e-06
74	2.031473e-02	70	91	2.896011e+00	4.622967e-17	2.156386e-03	3.867301e-04	4.643704e+01	2.243598e-06
75	1.977738e-02	70	91	2.896653e+00	4.303352e-17	2.027247e-03	3.689667e-04	4.601278e+01	2.199884e-06
76	1.927137e-02	70	91	2.897265e+00	3.874886e-17	1.908059e-03	3.532754e-04	4.556950e+01	2.091428e-06
77	1.879402e-02	70	91	2.897848e+00	4.026223e-17	1.797916e-03	3.368892e-04	4.511034e+01	1.990600e-06
78	1.834294e-02	70	91	2.898404e+00	4.073183e-17	1.946780e-03	3.705573e-04	5.143935e+01	2.051868e-06
79	1.795496e-02	70	91	2.899010e+00	4.351449e-17	1.822987e-03	3.512236e-04	5.089170e+01	1.972540e-06
80	1.748253e-02	70	91	2.899585e+00	3.444204e-17	1.709588e-03	3.303551e-04	5.023789e+01	1.959848e-06
81	1.703779e-02	70	91	2.900129e+00	4.438181e-17	1.605285e-03	3.093104e-04	4.957269e+01	1.884210e-06
82	1.661887e-02	70	91	2.900644e+00	4.358208e-17	1.509203e-03	2.900079e-04	4.890071e+01	1.410200e-06
83	1.622360e-02	70	91	2.901133e+00	3.718136e-17	1.420562e-03	2.734237e-04	4.822525e+01	1.354506e-06
84	1.585002e-02	70	91	2.901596e+00	2.951070e-17	1.338663e-03	2.612777e-04	4.754900e+01	1.307318e-06
85	1.549638e-02	70	91	2.902035e+00	4.262520e-17	1.262883e-03	2.490797e-04	4.687422e+01	8.617236e-07
86	1.516115e-02	70	91	2.902452e+00	2.909744e-17	1.192664e-03	2.377913e-04	4.620281e+01	8.225895e-07
87	1.484292e-02	70	91	2.902848e+00	3.262872e-17	1.294892e-03	2.596611e-04	5.247291e+01	9.665419e-07
88	1.456834e-02	70	91	2.903278e+00	4.421213e-17	1.215309e-03	2.445770e-04	5.167471e+01	1.528806e-06
89	1.423269e-02	70	91	2.903682e+00	3.002183e-17	1.158734e-03	2.302144e-04	5.081238e+01	1.522439e-06
90	1.391526e-02	70	91	2.904064e+00	4.785701e-17	1.120552e-03	2.170064e-04	4.996107e+01	1.380657e-06
91	1.361492e-02	70	91	2.904423e+00	4.706976e-17	1.083934e-03	2.048211e-04	4.912314e+01	1.315823e-06
92	1.333033e-02	70	91	2.904761e+00	4.982881e-17	1.048829e-03	1.947950e-04	4.830005e+01	1.255283e-06
93	1.306026e-02	70	91	2.905080e+00	3.788125e-17	1.015184e-03	1.855037e-04	4.749284e+01	1.213742e-06
94	1.280365e-02	70	91	2.905381e+00	4.208391e-17	9.829397e-04	1.766991e-04	4.670224e+01	1.099163e-06
95	1.255950e-02	70	91	2.905664e+00	4.914612e-17	9.520392e-04	1.671058e-04	4.592873e+01	1.145618e-06
96	1.232692e-02	70	91	2.905931e+00	4.208357e-17	9.224242e-04	1.591638e-04	4.517263e+01	1.049329e-06
97	1.210512e-02	70	91	2.906182e+00	4.852325e-17	1.030675e-03	1.748997e-04	5.121100e+01	1.157267e-06
98	1.191436e-02	70	91	2.906453e+00	3.742087e-17	9.937791e-04	1.660231e-04	5.031573e+01	1.052734e-06
99	1.167851e-02	70	91	2.906705e+00	3.369977e-17	9.595523e-04	1.573717e-04	4.938993e+01	1.052038e-06
100	1.145437e-02	70	91	2.906940e+00	5.271342e-17	9.269007e-04	1.497935e-04	4.848879e+01	9.836016e-07